

Active Retrieval-Augmented Generation with Conflict-Fused Uncertainty Quantification

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Abstract

Active retrieval-augmented generation (RAG) triggers external knowledge retrieval during generation based on model-side uncertainty signals to support knowledge-intensive, multi-hop reasoning. However, existing methods often retrieve only after producing a complete answer, failing to surface and fill information gaps in time; moreover, relying on a single internal signal as the trigger cannot adequately capture the multifaceted nature of uncertainty. We therefore propose a conflict-aware active RAG framework. We first decompose complex questions into a sequence of step-level sub-problems. At each step, we quantify local distributional uncertainty via a sliding-window peak token entropy, and estimate cross-sample consensus via the variation ratio computed over multiple Monte Carlo samples. After calibrating both signals onto a probabilistic scale, we quantify their conflict using a symmetric, bounded divergence over Bernoulli parameters, and fuse the three quantities into a single uncertainty score that gates retrieval. Experiments demonstrate the effectiveness of our framework.

CCS Concepts

- Computing methodologies → Natural language generation;
- Information systems → Question answering.

Keywords

Large Language Model, Active Retrieval-Augmented Generation, Uncertainty Estimation, Multi-hop Reasoning QA

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Chain-of-Thought Error Propagation Low Entropy → No Retrieval

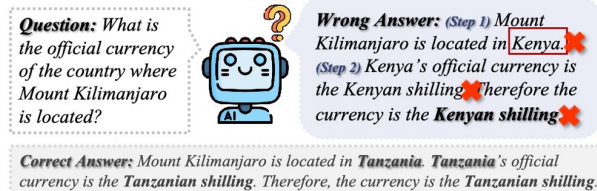


Figure 1: A typical failure case where a confidently wrong intermediate step poisons the final multi-hop answer.

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1 Introduction

Large language models (LLMs) have achieved impressive performance on a wide range of NLP tasks [1], but their parametric knowledge is inevitably incomplete and often outdated [5]. Retrieval-augmented generation (RAG) addresses this limitation by conditioning the model on external documents from a large corpus such as Wikipedia [14]. For knowledge-intensive, multi-hop question answering (QA), recent work further shows that interleaving retrieval and generation can ground intermediate reasoning steps in retrieved evidence [17, 18].

A key challenge in RAG is therefore not only *how* to retrieve, but also *when*. Classical RAG frameworks such as retrieval-augmented generation [8] and FiD [6] follow a retrieve-then-generate pipeline, performing a single retrieval pass before decoding the whole answer or using fixed multi-round schedules that are blind to the model's confidence [16]. More recent *active* RAG methods make the timing decision by monitoring a single internal signal. For example, FLARE [7] tracks token log-probabilities and entropy along the output prefix and triggers sentence-level retrieval when it detects low-confidence spans, while DRAGIN [15] estimates attention-based uncertainty over retrieved passages and dynamically requests more evidence when the attention distribution becomes diffuse. These approaches demonstrate that uncertainty-aware scheduling can improve RAG, but they still suffer two structural limitations: (i) they essentially rely on a *single* view of uncertainty (e.g., token

likelihoods or attention entropy summarized into one scalar), and (ii) their signals are computed mainly at the token or sentence level, which can be misaligned with the step-wise reasoning structure of multi-hop QA.

Given a multi-hop question in Figure 1 about the currency of the country where Mount Kilimanjaro is located, the LLM first (incorrectly) states that the mountain is in Kenya, then correctly recalls Kenya’s official currency, and finally concludes that the answer is “Kenyan shilling”. Because the model is locally confident in both steps, token-level entropy remains low in this case, and entropy-based triggers fail to fire on the crucial first hop. The early factual mistake silently propagates along the chain-of-thought and poisons the final answer. Post-hoc retrieval on the complete answer is therefore too late to repair the reasoning trajectory. This example highlights a broader issue: treating uncertainty as a one-dimensional, token-level quantity leaves substantial blind spots, especially for confidently wrong intermediate steps.

Therefore, we propose a parameter-efficient active RAG framework, *Active Retrieval-Augmented Generation with Conflict-Fused Uncertainty Quantification* (CFU-RAG), which keeps the backbone LLM frozen. We decompose complex questions into step-level sub-problems and, at each inference step, decide whether to trigger retrieval based on an uncertainty signal. Specifically, we use token-level entropy to characterize the local confidence of the current step, apply Monte Carlo decoding to capture cross-sample disagreement for the same step under stochastic sampling, and explicitly define a conflict term that measures the discrepancy between these two views. These three signals are fused into a single step-level uncertainty score that gates retrieval. The conflict term amplifies the overall uncertainty when the two views issue inconsistent judgments about the reliability of the step. Across both 2WikiMulti-hopQA and HotpotQA, CFU-RAG consistently improves EM and F1 over strong RAG baselines.

2 Methodology

We propose an active retrieval-augmented generation framework (CFU-RAG) in which conflict-aware step-level uncertainty triggers retrieval of external evidence. Specifically, our framework consists of (1) segmenting answers into simple reasoning steps and deriving a conflict-aware step-level uncertainty signal by combining multiple uncertainty estimates, and (2) passing this signal to an uncertainty-driven retrieval gating mechanism that decides when to query external evidence and refine the reasoning steps.

2.1 Step-wise Uncertainty Quantification

We use a fixed few-shot prefix that demonstrates how a question is decomposed into several intermediate reasoning steps followed by a short final answer, and prompt the LLM to follow this pattern for the current question. Conditioned on this prompt (few-shot exemplars, optional *Searched Information*, and the partial answer block), CFU-RAG generates the answer step by step. After each newly produced reasoning step, we compute two complementary uncertainty signals: a cross-sample disagreement score S from Monte Carlo (MC) decoding, and a local distributional score A from token-level entropy along the current decode path. These signals are later calibrated, combined with a conflict term, and used to gate

retrieval. Formally, let x denote the current anchor prompt and $y = (y_1, \dots, y_T)$ the continuation tokens of the segment; at each position t , the backbone LLM f_θ induces a predictive distribution $p_\theta(\cdot | x, y_{<t})$ over the vocabulary $\mathcal{V}$.

Cross-Sample Disagreement. To capture epistemic uncertainty manifested as *divergent* but individually confident continuations, we perform MC decoding from the same anchor x [2] and measure disagreement at the segment level. We draw K independent continuations,

$$\mathcal{Y} = \{y^{(1)}, y^{(2)}, \dots, y^{(K)}\},$$

and apply a light normalization function $g(\cdot)$ (lowercasing, punctuation removal, whitespace collapsing) to obtain $\tilde{y}^{(i)} = g(y^{(i)})$. Let $\mathcal{U}$ be the set of distinct normalized candidates, and define

$$n(u) = |\{i : \tilde{y}^{(i)} = u\}|, \quad \hat{p}(u) = \frac{n(u)}{K}, \quad (1)$$

for each $u \in \mathcal{U}$. We then use the *variation ratio* as a disagreement measure:

$$S = 1 - \max_{u \in \mathcal{U}} \hat{p}(u), \quad (2)$$

so that $S = 0$ when all MC decodes agree and larger values indicate higher cross-sample disagreement.

Local Distributional Uncertainty. We also quantify local distributional uncertainty from the token-level predictive distribution along a single decode path. At position t , we write

$$\pi_t(v) = p_\theta(v | x, y_{<t}), \quad v \in \mathcal{V},$$

and apply the same temperature and nucleus (top- p) truncation as used for decoding [4], yielding a truncated distribution $\tilde{\pi}_t$ supported on $\mathcal{V}_t \subseteq \mathcal{V}$. We then define a normalized token entropy

$$h_t = -\frac{1}{\log |\mathcal{V}_t|} \sum_{v \in \mathcal{V}_t} \tilde{\pi}_t(v) \log \tilde{\pi}_t(v) \in [0, 1]. \quad (3)$$

During MC decoding, each rollout i produces a sequence of content tokens for the segment together with entropies

$$\mathbf{h}^{(i)} = (h_1^{(i)}, h_2^{(i)}, \dots, h_{T_i}^{(i)}),$$

where T_i is the number of content tokens in rollout i and we discard stopwords and pure punctuation. Given a window length L in content tokens, we define an *anywhere- L sliding-window peak* for each rollout (setting $A^{(i)} = 0$ when $T_i = 0$):

$$A^{(i)} = \max_{1 \leq j \leq \max(1, T_i - L + 1)} \frac{1}{\min(L, T_i)} \sum_{t=j}^{j+\min(L, T_i)-1} h_t^{(i)}, \quad (4)$$

i.e., we take the maximum average entropy over all windows of length L , falling back to the entire segment if it is shorter than L . To aggregate across the K rollouts and reduce the effect of outliers, we use a high quantile

$$A = q_{0.8}(\{A^{(i)}\}_{i=1}^K), \quad (5)$$

the empirical 80-th percentile of the per-rollout peaks, which yields a local distributional uncertainty score $A \in [0, 1]$.

Calibration and Conflict Score. Although A and S both lie in $[0, 1]$, their scales and semantics differ. We therefore map each to an approximate probability that the current segment is unreliable using one-dimensional calibration on a development set [12]. Concretely, we collect tuples (A, y) and (S, y) , where $y \in \{0, 1\}$ indicates

whether the segment requires additional retrieval or is factually incorrect, and learn calibration functions

$$p_A = C_A(A), \quad p_S = C_S(S), \quad (6)$$

via Platt scaling [12]. Note that this calibration is the only step requiring data, and it is computationally negligible compared to backbone fine-tuning, preserving the training-free nature of the LLM itself. After calibration, $p_A, p_S \in [0, 1]$ can be interpreted as two probabilistic estimates of the same binary event.

To capture the *conflict* between these two views, we treat p_A and p_S as parameters of Bernoulli distributions and measure the discrepancy between $\text{Bern}(p_A)$ and $\text{Bern}(p_S)$. Let JSD denote the Jensen–Shannon divergence normalized to $[0, 1]$ by dividing by $\log 2$ [9]. We define a symmetric, bounded divergence

$$C = \sqrt{\text{JSD}(\text{Bern}(p_A) \parallel \text{Bern}(p_S))} \in [0, 1], \quad (7)$$

Finally, drawing inspiration from opinion aggregation in multi-view learning [11], we obtain a conflict-fused uncertainty score by linearly combining the calibrated probabilities and their conflict:

$$U = \alpha A + \beta S + \gamma C, \quad \alpha, \beta, \gamma \geq 0, \quad \alpha + \beta + \gamma = 1. \quad (8)$$

In the subsequent section, we use U together with a dual-high fallback on $\min(A, S)$ to gate retrieval decisions during decoding.

2.2 Uncertainty-Gated Active Retrieval

For a question q , at step k we maintain the current reasoning trace $\{s_1, \dots, s_{k-1}\}$ and the set of retrieved passages $\mathcal{D}_k$. Together with the fixed few-shot prefix, these form the anchor prompt x_k , from which the model generates a new step candidate s_k (or a final answer clause) as described in Section 2.1. We then run the uncertainty estimator on this segment to obtain (A_k, S_k, C_k, U_k) .

Dual gating rule. We use the fused uncertainty score U_k as the primary trigger for retrieval [10], but also add a “both-high” backstop to avoid averaging away cases where both views are uncertain. We write $\mathbb{I}[\cdot]$ for the indicator function, which equals 1 if the condition holds and 0 otherwise. Let T be the main threshold and T_{both} the backstop threshold. We trigger retrieval at step k whenever

$$g_k = \mathbb{I}[U_k \geq T \vee \min(A_k, S_k) \geq T_{\text{both}}], \quad (9)$$

subject to a global budget of at most B retrieval rounds per question.

Retrieval and step progression. When the gate fires ($g_k = 1$) and budget remains, we construct a retrieval query by combining the original question q with the most recent non-empty steps in the answer block, and issue this query to a BM25 retriever [13] over Wikipedia to obtain a small set of passages. These passages are appended to the “Searched Information” block and the current step s_k is regenerated from the same anchor position under the updated context, yielding refreshed uncertainty scores (A_k, S_k, C_k, U_k) . Once s_k is finalized (with or without retrieval), it is appended to the reasoning trace and the controller decides whether to generate another intermediate step or stop. In practice we use a simple hop cap K_{max} together with a lightweight heuristic that checks whether the model has already produced a concluding phrase (e.g., “So the answer is ...”); the final answer is then taken from the clause following this phrase and treated as the last segment. Overall, the controller is training-free with respect to the backbone LLM.

3 Experiments

3.1 Experimental Setup

Dataset and backbone. We evaluate on 2WikiMultihopQA [3] and HotpotQA [17], two multi-hop QA benchmarks in which each question requires combining evidence from at least two Wikipedia articles and is accompanied by annotated reasoning paths. We follow the standard data splits and evaluation protocol, reporting performance on the development set. Unless otherwise stated, all methods use the same Llama2-7b-chat backbone without any additional fine-tuning; our approach is thus training-free with respect to the backbone.

Retrieval and implementation. We follow DRAGIN [15] and build a BM25 index over the same English Wikipedia snapshot. For all RAG variants, each retrieval call returns at most n passages (typically $n=5$), which are prepended as a *Searched Information* block in the prompt. For CFU-RAG, we use the MC decoding and uncertainty estimation described in Section 2.2. We set the number of Monte Carlo samples $K = 5$ to balance performance and latency. The fusion weights are fixed to $\alpha = 0.4$, $\beta = 0.4$, and $\gamma = 0.2$ in the conflict-fused uncertainty score U , and gating thresholds set to $T = 0.7$ and $T_{\text{both}} = 0.4$. All experiments are conducted on a single NVIDIA A100 GPU.

Baselines. We compare CFU-RAG against several RAG variants implemented under a unified framework. *wo-RAG* is vanilla Llama2-7b-chat used without any retrieval. *SR-RAG* is a single-round RAG variant that retrieves once given the question and then generates the answer with all retrieved passages at once. *FL-RAG* is a fixed-length active RAG variant that triggers retrieval after a fixed number of generated tokens. *FS-RAG* is a fixed-step active RAG variant that triggers retrieval after a fixed number of reasoning steps. *FLARE* [7] is an active RAG variant that uses token-level uncertainty surrogates to decide when to retrieve. *DRAGIN* [15] uses attention-entropy-based gating for dynamic retrieval.

Method	2WikiMultihopQA		HotpotQA	
	EM	F1	EM	F1
wo-RAG	0.158	0.236	0.174	0.257
SR-RAG	0.168	0.251	0.164	0.265
FL-RAG	0.138	0.191	0.156	0.242
FS-RAG	0.156	0.203	0.208	0.293
FLARE	0.168	0.248	0.122	0.196
DRAGIN	0.224	0.290	0.234	0.334
CFU-RAG (ours)	0.276	0.364	0.261	0.352

Table 1: Main results on 2WikiMultihopQA and HotpotQA.

3.2 Comparison with SOTA

Table 1 reports exact match (EM) and token-level F1 on both 2WikiMultihopQA and HotpotQA. Overall, all active RAG variants outperform the non-retrieval baseline (wo-RAG) on at least one metric, confirming the benefit of incorporating external evidence. Among the baselines, DRAGIN consistently achieves the strongest performance, improving over single-round (SR-RAG) and fixed-schedule (FL-RAG, FS-RAG) retrieval, and substantially outperforming FLARE, which relies solely on token-level uncertainty surrogates.

CFU-RAG further improves upon DRAGIN on both datasets, achieving the best EM/F1 under the same backbone. On 2WikiMultihopQA, CFU-RAG yields a noticeable gain over DRAGIN (from 0.224/0.290 to 0.276/0.364 in EM/F1), and on HotpotQA it also improves EM and F1 (from 0.234/0.334 to 0.261/0.352). These consistent gains across datasets suggest that conflict-fused step-wise uncertainty provides a more reliable retrieval trigger than single-view, token-level or attention-based criteria.

3.3 Ablation on Uncertainty Fusion

To understand the contribution of the two uncertainty views and their fusion, we ablate the gating rule in CFU-RAG while keeping all other components fixed. We consider three variants: *A-only*, which triggers retrieval solely based on the local distributional uncertainty A_k ; *S-only*, which uses only the cross-sample disagreement S_k ; and *Fusion*, which adopts the full dual gate with the conflict-fused score U_k and the both-high backstop on $\min(A_k, S_k)$. Table 2 summarizes the results on 2WikiMultihopQA.

Gating variant	EM	F1
A-only (local entropy)	0.250	0.313
S-only (VR disagreement)	0.268	0.360
Fusion (CFU-RAG, full)	0.276	0.364

Table 2: Ablation on the uncertainty gating rule on 2WikiMultihopQA.

Several trends emerge. First, S-only clearly outperforms A-only (EM 0.268 vs. 0.250, F1 0.360 vs. 0.313), indicating that cross-sample disagreement is a stronger standalone indicator of problematic reasoning steps than local token entropy. This matches our intuition that many failure cases arise from *confidently wrong* but *divergent* continuations that keep A_k low while producing large variability across MC samples. Second, the Fusion variant achieves the best performance (EM 0.276, F1 0.364), improving over both A-only and S-only. Although the gains over S-only are numerically modest, they are consistent and come purely from changing the gating policy.

Qualitatively, we observe that A-only tends to miss cases where the model follows several mutually inconsistent but individually low-entropy trajectories, leading to under-triggering of retrieval on early hops; in contrast, S-only tends to be noisy on benign steps where surface realizations differ but the underlying reasoning is aligned. The conflict-fused gate combines the strengths of both views, while the both-high backstop prevents averaging effects from suppressing retrieval when *both* A_k and S_k are high. This explains why the Fusion variant is more robust overall in the ablation results.

4 Conclusion

We studied when to trigger retrieval in active RAG for multi-hop QA. Existing methods mainly rely on a single, token-level uncertainty signal and often miss confidently wrong intermediate steps. We proposed CFU-RAG, a training-free framework that operates at the step level and fuses local entropy, cross-sample disagreement, and their conflict into a single uncertainty score for gating retrieval. In experiments on 2WikiMultihopQA and HotpotQA with a shared

Llama2-7b-chat backbone and BM25 retriever, CFU-RAG consistently improves EM/F1 over strong RAG baselines, and ablations show that conflict-fused uncertainty yields more reliable gating than either uncertainty view alone.

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References

- [1] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [2] Yarin Gal and Zoubin Ghahramani. 2016. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *international conference on machine learning*. PMLR, 1050–1059.
- [3] Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. *arXiv preprint arXiv:2011.01060* (2020).
- [4] Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2019. The curious case of neural text degeneration. *arXiv preprint arXiv:1904.09751* (2019).
- [5] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. 2025. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems* 43, 2 (2025), 1–55.
- [6] Gautier Izacard and Edouard Grave. 2021. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th conference of the european chapter of the association for computational linguistics: main volume*. 874–880.
- [7] Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 7969–7992.
- [8] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.
- [9] Jianhua Lin. 2002. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information theory* 37, 1 (2002), 145–151.
- [10] Wei Liu, Yufei Chen, and Xiaodong Yue. 2025. Enhancing Multi-View Classification Reliability with Adaptive Rejection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 18969–18977.
- [11] Wei Liu, Xiaodong Yue, Yufei Chen, and Thierry Denoeux. 2022. Trusted multi-view deep learning with opinion aggregation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 7585–7593.
- [12] John Platt et al. 1999. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in large margin classifiers* 10, 3 (1999), 61–74.
- [13] Stephen Robertson, Hugo Zaragoza, et al. 2009. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends® in Information Retrieval* 3, 4 (2009), 333–389.
- [14] Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston. 2021. Retrieval augmentation reduces hallucination in conversation. *arXiv preprint arXiv:2104.07567* (2021).
- [15] Weihang Su, Yichen Tang, Qingyao Ai, Zhijiang Wu, and Yiqun Liu. 2024. DRAGIN: dynamic retrieval augmented generation based on the information needs of large language models. *arXiv preprint arXiv:2403.10081* (2024).
- [16] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*. 10014–10037.
- [17] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*. 2369–2380.
- [18] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R Narasimhan, and Yuan Cao. 2022. React: Synergizing reasoning and acting in language models. In *The eleventh international conference on learning representations*.